Enterprise Blockchain Case Studies

TypeScript repository for enterprise blockchain design, protocol mapping, and integration boundaries.

Scope

- Case studies tied to operating problems
- Protocol-specific transaction models
- SDK-oriented integration patterns
- Off-chain cryptographic primitives (MPC, HSM, PQC)

Repository Structure

Area	Purpose
`modules/`	Domain logic, protocol adapters, and integration clients
`examples/`	Runnable case studies and protocol projections
`contracts/`	Solidity source, Fabric chaincode, and ABI artifacts
`docs/`	Research, architecture notes, and presentation material
`skills/`	AI skill files for coding assistants and agents

Operating Problems Covered

1. Food recall response
2. Consortium order sharing
3. Hospital staffing clearance
4. Aid voucher reconciliation

These scenarios require provenance, privacy, cross-organization coordination, or settlement controls.

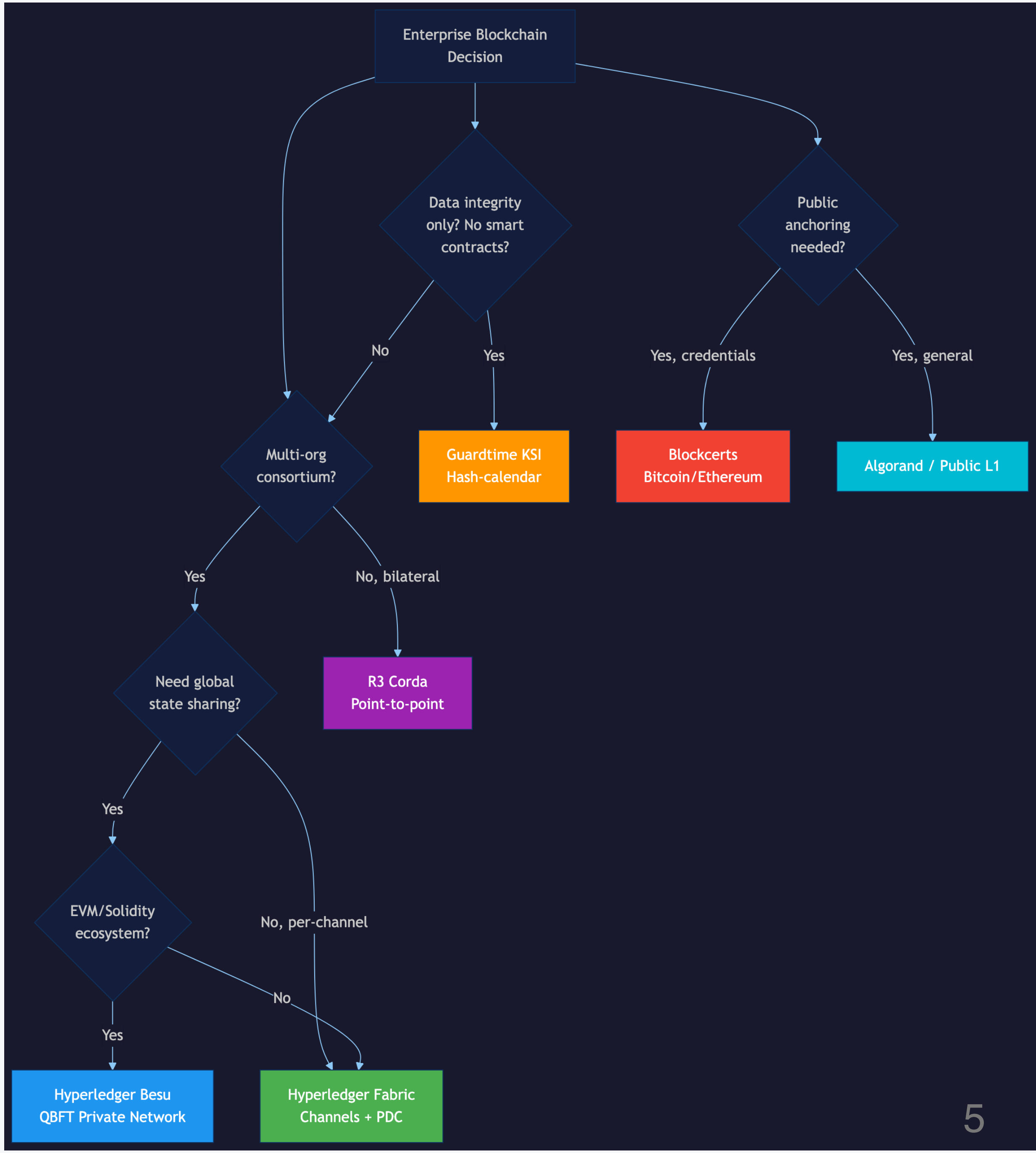
Off-Chain Cryptographic Patterns

Pattern	Purpose
MPC	Joint computation without disclosing individual inputs
HSM	Hardware-protected key custody and signature provenance
PQC	Post-quantum cryptography for long-term confidentiality

These complement ledger-based scenarios with cryptographic guarantees.

Architecture Model

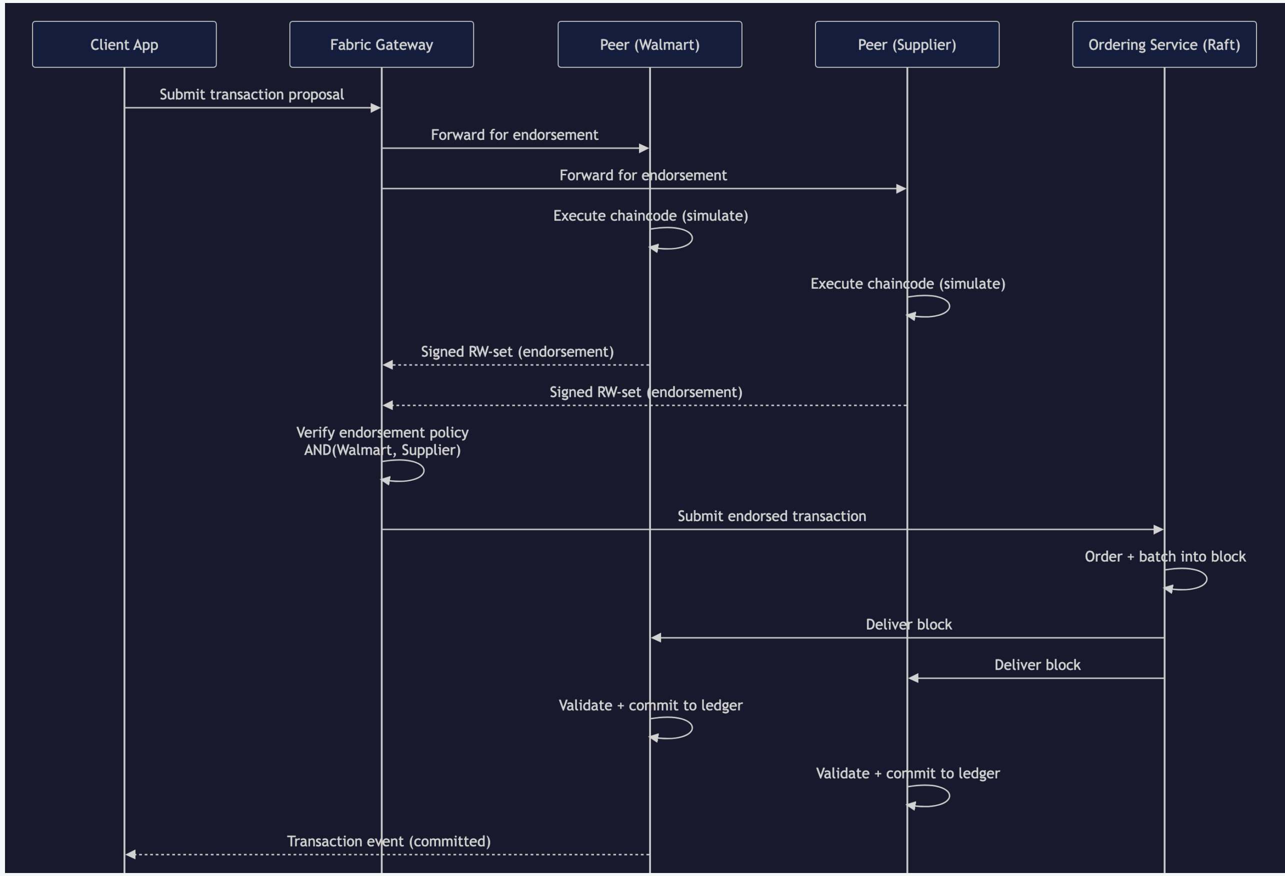
- Domain modules hold the business rules.
- Protocol adapters capture platform semantics.
- Integration clients shape runtime requests.
- Environment configuration supplies operational inputs.



Food Recall Response

Why it matters

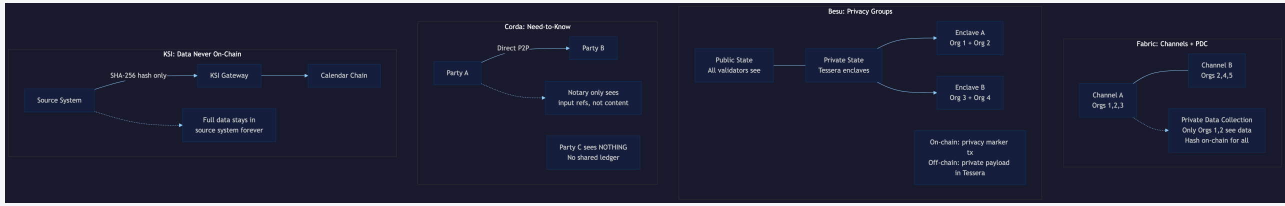
- Rapid impact analysis during a recall
- Shared traceability across supplier, carrier, and retailer
- Deterministic commit model for operational response



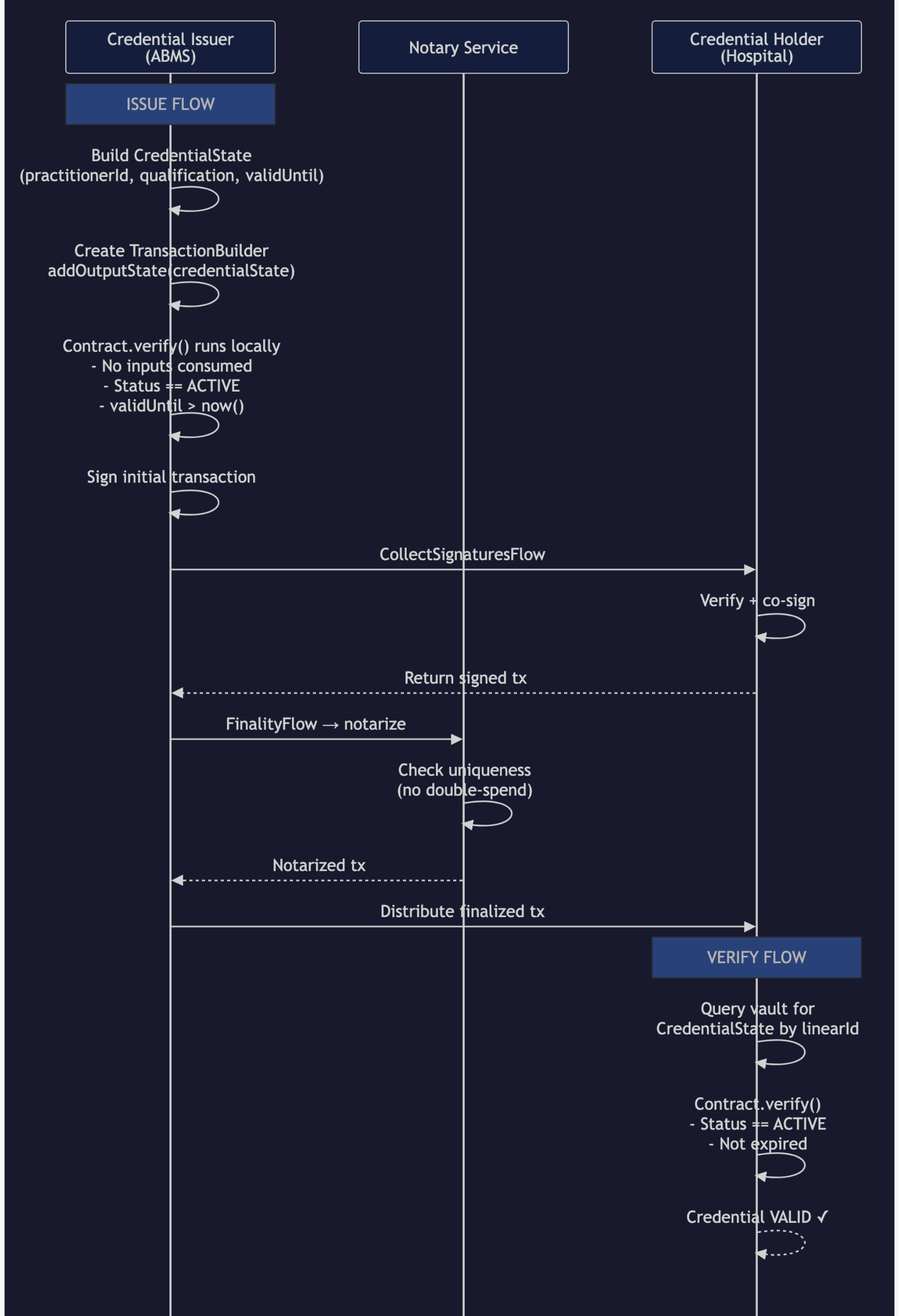
Selective Disclosure

Key point

The same commercial record can be anchored once and disclosed differently to logistics, banking, regulatory, and supplier audiences.



Credential Verification



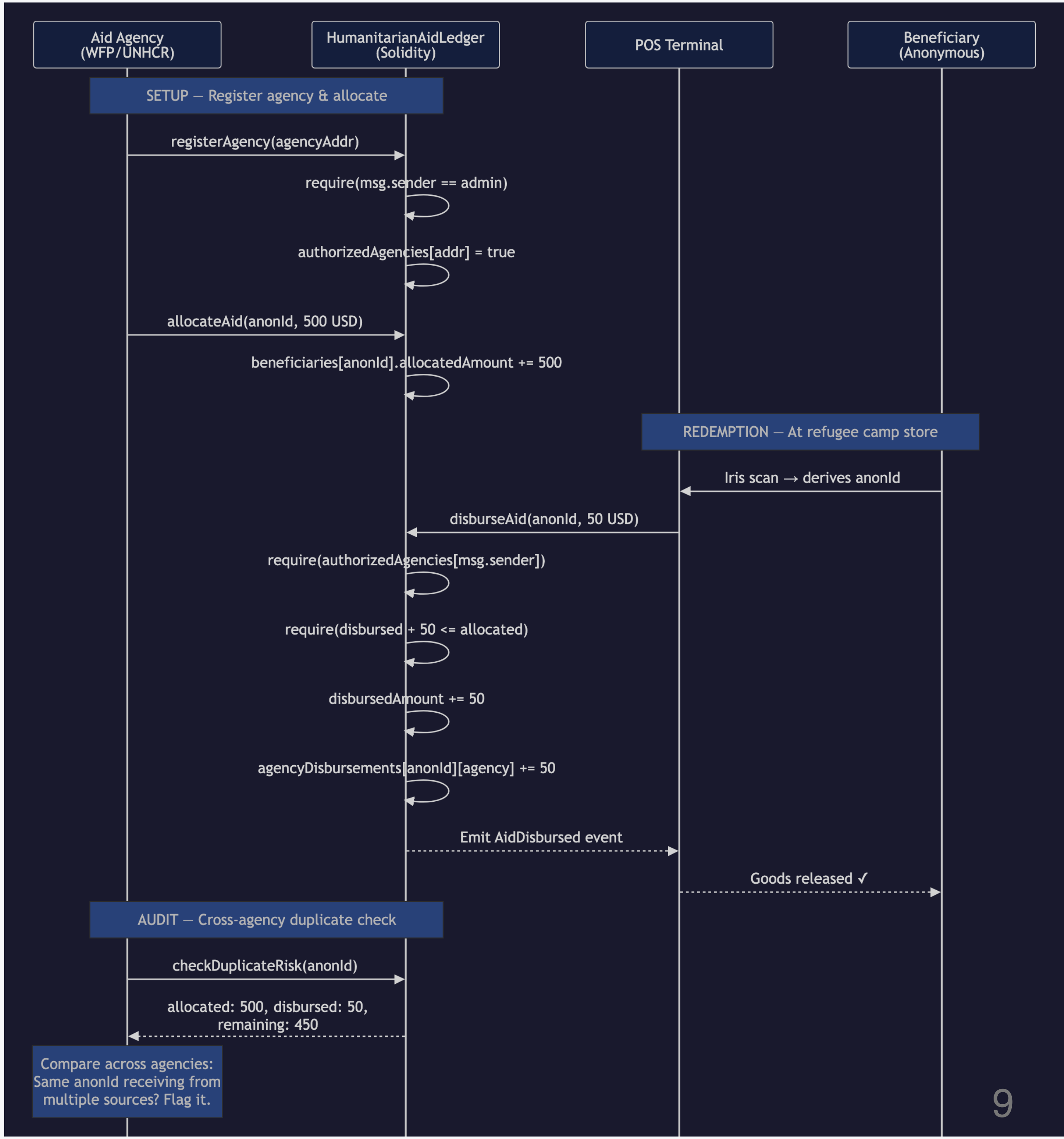
Key point

Need-to-know distribution is often a better fit than broad ledger replication for regulated staffing workflows.

Reconciliation And Controls

Key point

Shared ledgers matter when multiple organizations need one settlement view without delegating control to a single operator.



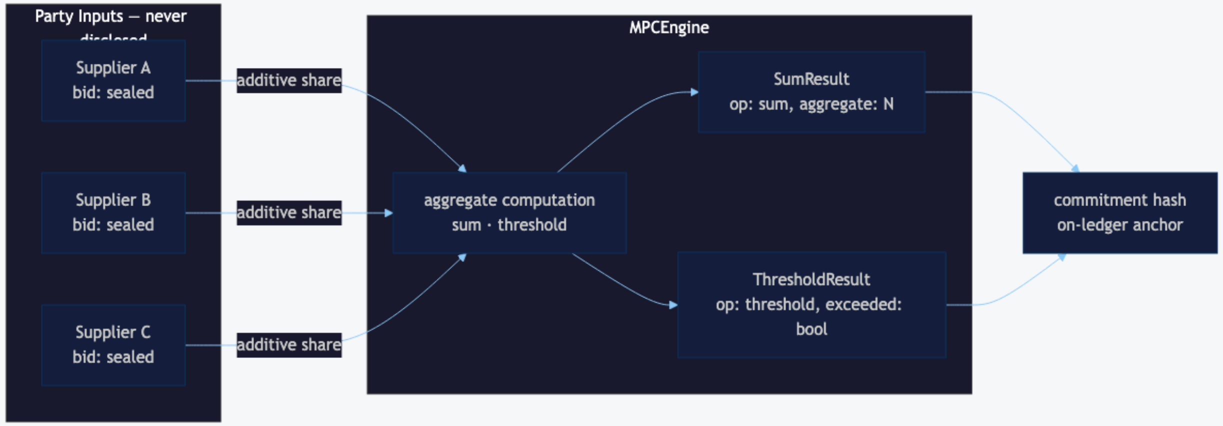
MPC: Secret Sharing

Key point

Parties compute a joint result without any participant revealing their individual input.

Repository examples:

- `mpc-sealed-bid-auction` — additive secret sharing
- `mpc-joint-risk-analysis` — cross-institution credit threshold



MPC: Threshold Schemes

Shamir Secret Sharing (k-of-n)

Distribute a secret so that any k shares reconstruct it, but k–1 shares reveal nothing.

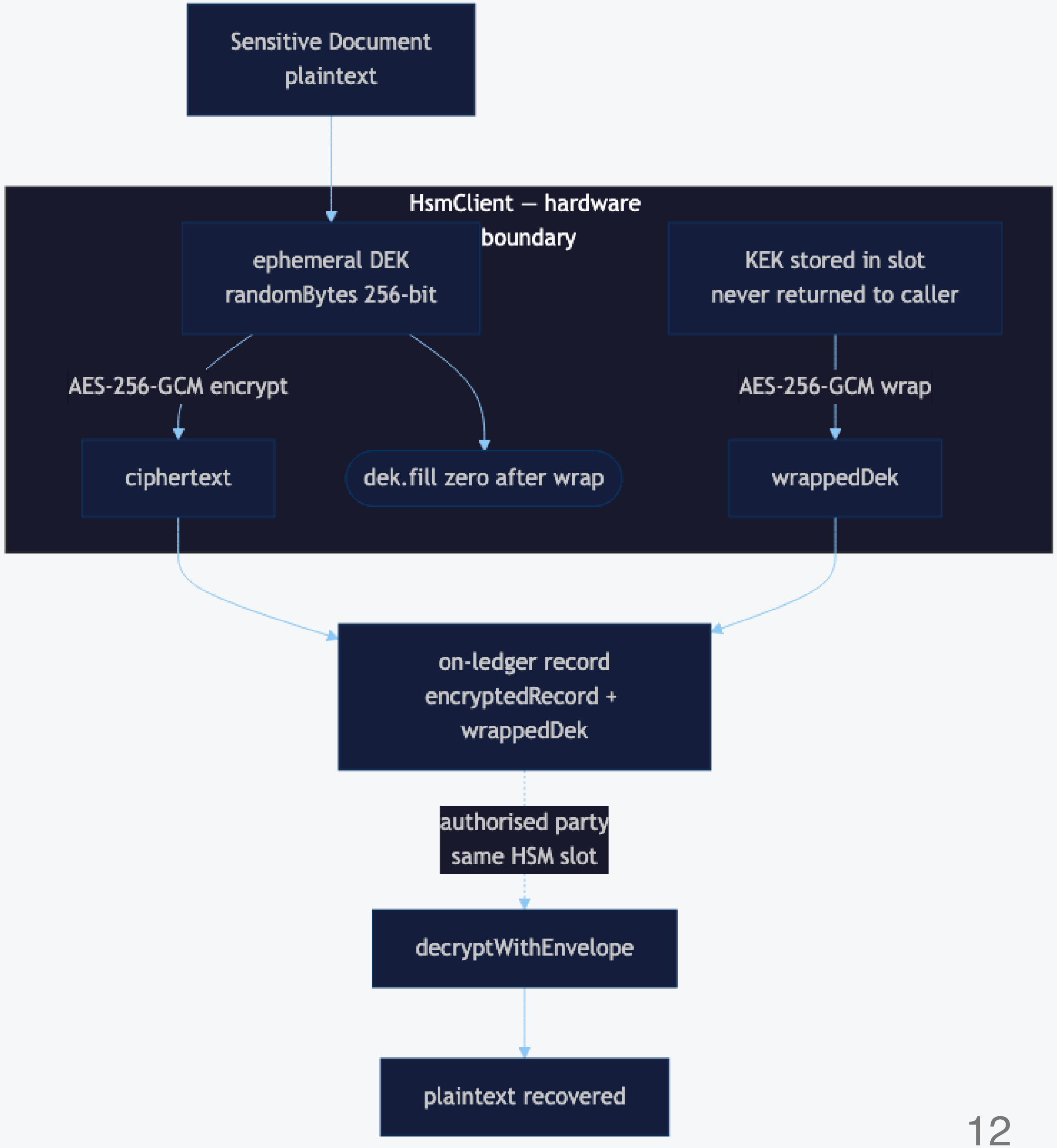
Repository example:

- `quantum-resistant-key-sharing` — 3-of-5 Shamir threshold with hash-ladder anchoring for post-quantum auditability

HSM Key Management

Key point

Private keys and raw symmetric material never leave the HSM boundary. Only signatures, public keys, and wrapped DEKs appear on-chain.



HSM Examples

Example	Pattern
`hsm-transaction-signing`	EC P-256 trade order signing with HSM attestation
`hsm-key-ceremony`	3-of-5 Shamir custodianship + HSM-signed certificate
`hsm-envelope-encryption`	DEK/KEK pattern for on-ledger document confidentiality

Post-Quantum Cryptography

NIST FIPS 203 & 204 (2024)

Standard	Algorithm	Purpose
FIPS 203	ML-KEM (Kyber)	Key encapsulation mechanism
FIPS 204	ML-DSA (Dilithium)	Digital signature algorithm

These provide quantum-resistant replacements for RSA, ECDH, and ECDSA.

Kyber KEM Key Exchange

ML-KEM (FIPS 203)

- Lattice-based key encapsulation
- Parameter sets: ML-KEM-512, ML-KEM-768, ML-KEM-1024
- Shared secret derived via encapsulation/decapsulation

Repository example:

- `kyber-kem-key-exchange` — Full ML-KEM roundtrip with audit records

Hybrid KEM Settlement

X25519 + ML-KEM-768

Combines classical (X25519) and post-quantum (ML-KEM) key exchange for defense-in-depth.

- HKDF combines both shared secrets
- Provides security even if one primitive breaks
- Backward compatibility with classical systems

Repository example:

- `hybrid-kem-settlement` — Settlement channel with hybrid key agreement

Quantum-Safe Payment Flow

End-to-end post-quantum security

1. **Key ceremony** — Hybrid KEM key pairs (X25519 + ML-KEM)
2. **Signing** — ML-DSA-65 digital signatures (FIPS 204)
3. **Encryption** — AES-256-GCM with hybrid-derived key
4. **Authorization** — 3-of-3 MPC threshold settlement

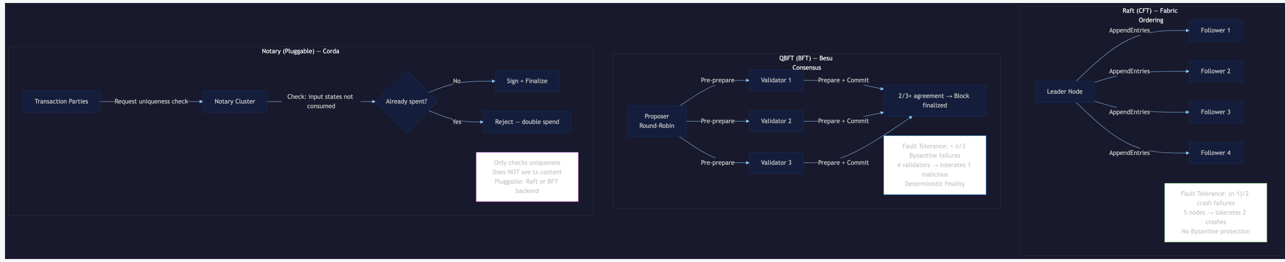
Repository example:

- `quantum-safe-payment` — Full FX settlement with PQC primitives

Platform Fit

Decision criteria in this repository:

- Governance model
- Privacy boundary
- Finality model
- Integration surface



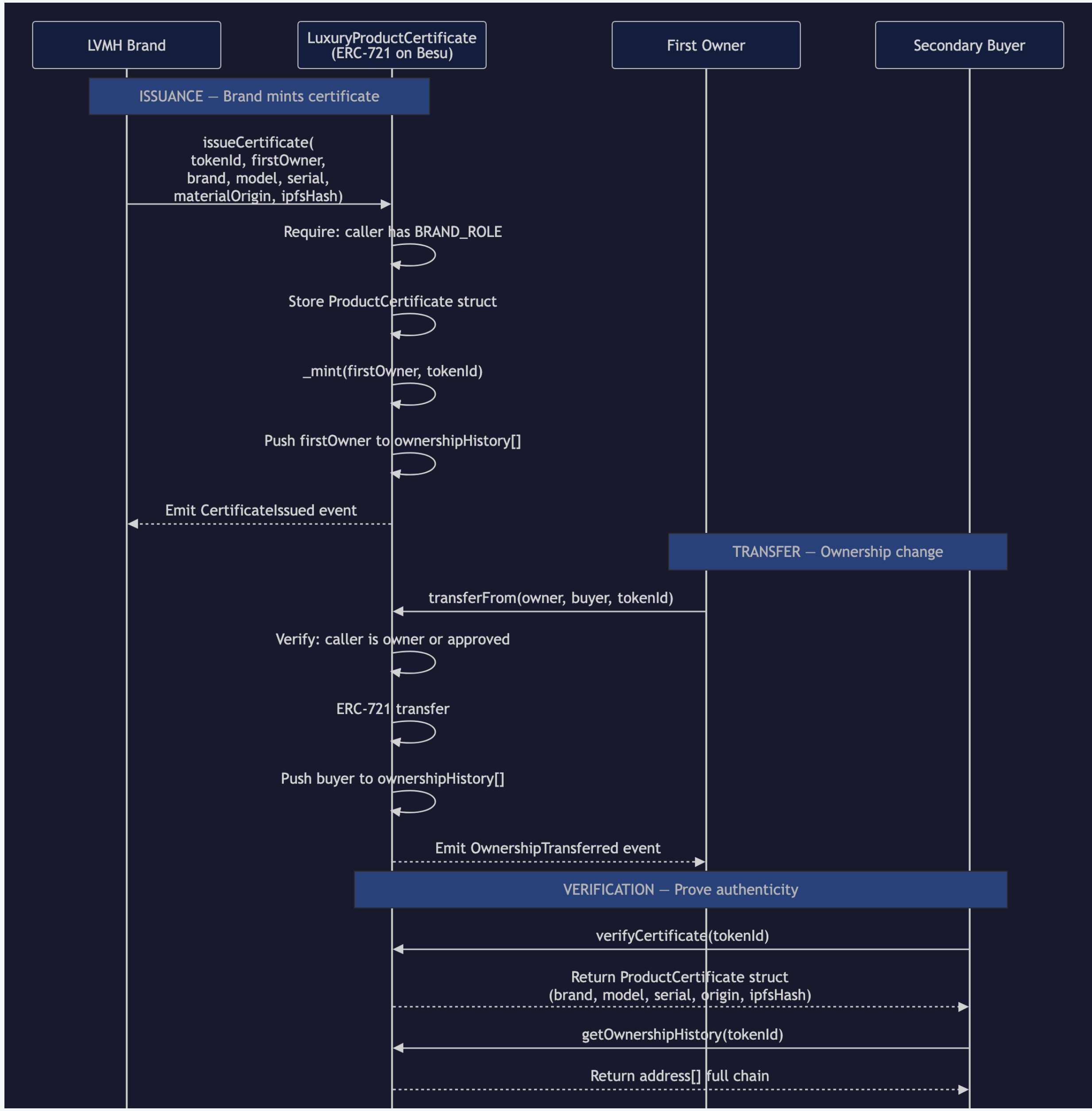
From Domain Logic To Runtime

1. Domain module defines the business rule.
2. Protocol adapter maps the event into a platform-specific transaction.
3. Integration client shapes the request for the runtime boundary.
4. Environment configuration supplies credentials and endpoints.

Besu Path

Repository assets:

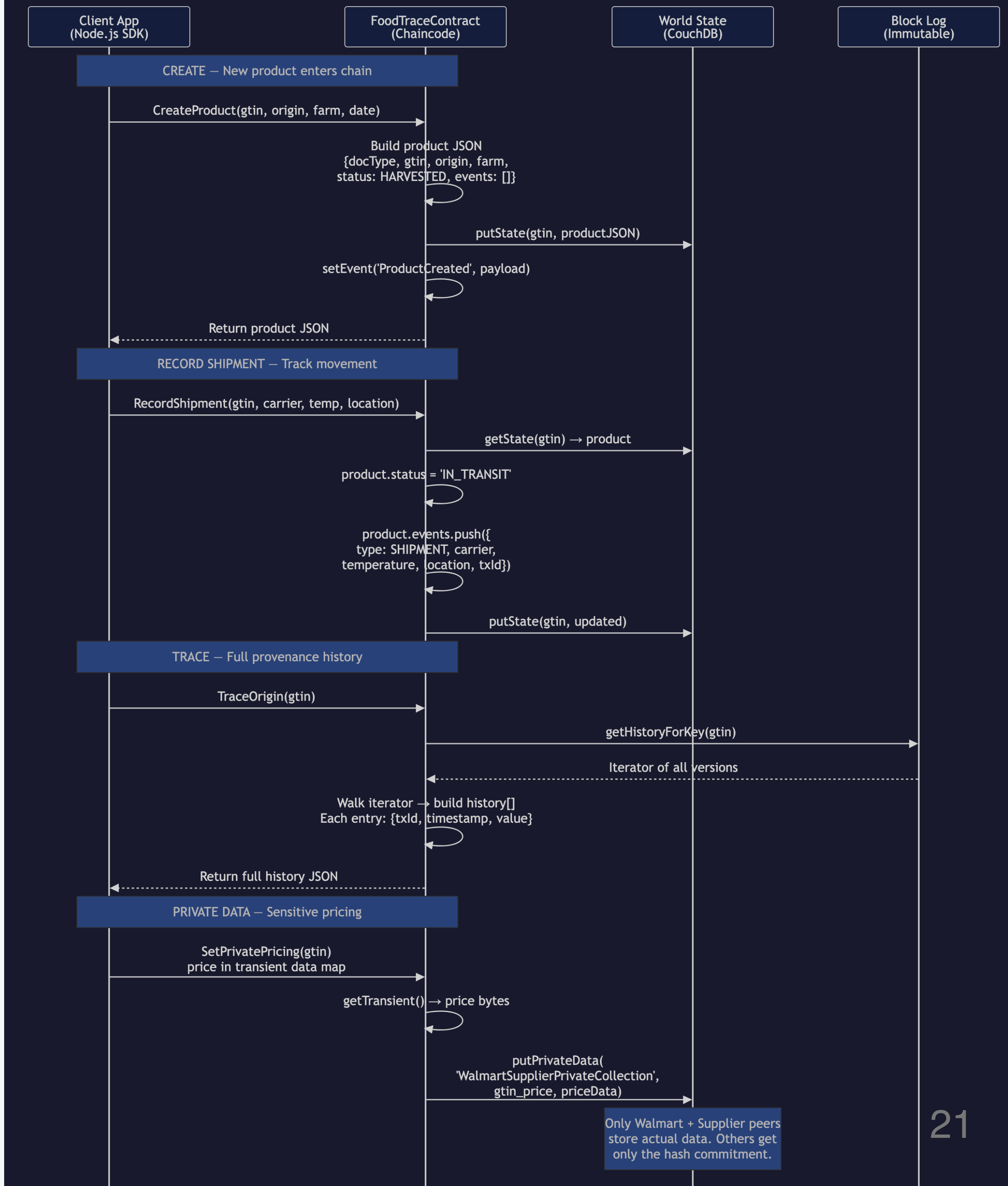
- Solidity source in `contracts/solidity/src/`
- ABI artifacts in `contracts/`
- `ethers` integration in `modules/integrations/besu-client/`



Fabric Path

Repository assets:

- Chaincode in `contracts/fabric/`
- Proposal builders in `modules/protocols/fabric/`
- Gateway integration in `modules/integrations/fabric-gateway/`



Corda Path

TypeScript is the integration layer, not the CorDapp runtime.

Repository assets:

- Kotlin contracts in `contracts/corda/`
- Flow projection in `modules/protocols/corda/`
- Gateway builder in `modules/integrations/corda-gateway/`

Quick Start

```
npm install
npm run verify
npm run demo:adapters
npm run demo:integrations
```

Case Study Examples

```
npm run example:food-recall
npm run example:order-sharing
npm run example:staffing-clearance
npm run example:aid-reconciliation
```

Protocol Projection Examples

```
npm run example:fabric-projection
npm run example:besu-projection
npm run example:corda-projection
npm run example:fabric-gateway
npm run example:besu-ethers
npm run example:corda-rest
```

MPC & Threshold Examples

```
npm run example:mpc-auction
npm run example:mpc-risk-analysis
npm run example:quantum-key-sharing
```

HSM Examples

```
npm run example:hsm-tx-signing
npm run example:hsm-key-ceremony
npm run example:hsm-envelope-encryption
```

Post-Quantum Examples

```
npm run example:kyber-kem
npm run example:hybrid-kem
npm run example:quantum-safe-payment
```

Navigation Guide

Goal	Start Here
Understand a scenario	`examples/` folder
Read domain logic	`modules/` folder
Review protocol shapes	`modules/protocols/`
See integration patterns	`modules/integrations/`
Study architecture	`docs/architecture/`

Summary

- **4** operating problems with domain modules
- **3** protocol adapters (Fabric, Besu, Corda)
- **3** integration clients (Gateway, ethers, REST)
- **6** MPC/HSM/PQC cryptographic examples

All examples pass `npm run verify` and run offline.